6.7 Modeling with Exponential and Logarithmic Functions

Name _____ Period ___ Date _____

Algebra II 6.7 · Student sheet

I can…

- tell whether a table shows a linear function, an exponential function, or neither.
- write an exponential function through two points.
- use technology to find an exponential model and make predictions.

Vocabulary · fill in the missing word

Constant differences — equal _____ and equal differences in y: linear

Constant ratios — equal x-steps and equal ratios of y: _____

Exponential regression — a _____ or Desmos model $y = ab^x$ that best fits data

Part 1 · Linear or exponential?

For each table (x = 0, 1, 2, 3, 4), find the differences and the ratios of consecutive y-values.

y-values	Differences	Ratios	Linear, exponential, or neither?
2, 5, 8, 11, 14			
4, 12, 36, 108, 324			
1, 3, 7, 13, 21			

Part 2 · Notes and examples (fill in each note, then work the example)

Linear or exponential?: Constant _____ → linear. Constant _____ → exponential.

Example 1

Is the table linear, exponential, or neither?

x: 0 1 2 3 4

y: 5 10 20 40 80

Example 2

Is the table linear, exponential, or neither?

x: 0 1 2 3 4

y: 3 5 9 15 23

Two points: To find $y = ab^x$ through two points: substitute both points, then _____ one equation by the other to find b.

Example 3

Write an exponential function $y = ab^x$ whose graph passes through (1, 6) and (3, 54).

Example 4

Write an exponential function $y = ab^x$ whose graph passes through (2, 12) and (4, 3).

Exponential regression: In Desmos, exponential regression is typed as _____.

Example 5

A town's population (in hundreds) t years after 2020 is shown. Use exponential regression to find a model. Predict the population in 2026.

t:	0	1	2	3	4
y:	120	150	185	232	290

Part 3 · Partner practice

Partner 1

Is the table linear, exponential, or neither? If exponential, write the function.

x:	0	1	2	3
y:	2	6	18	54

Partner 2

Write an exponential function through (0, 4) and (2, 36).